Distributional Lie derivatives, smooth invariant densities, and zero entropy in dimensions at most three

Abstract

Let M be a compact connected smooth manifold and X a smooth vector field with flow g_t. We work with the canonical density convention for scalar distributions: distributions are continuous linear functionals on smooth densities, and the distributional Lie derivative is defined by duality. We study the hypothesis

$$\text{if } u \in \mathcal{D}'(M) \text{ and } \mathcal{L}_X u = f \in C^\infty(M), \text{ then } u \in C^\infty(M). \tag{H}$$

We prove the local flow-box structure of distributional solutions, the functional-analytic consequences of (H) concerning the distributional kernel and the closed range of X on smooth functions, and the precise dynamical exclusions obtained from singular distributional witnesses: trace-zero equilibria and periodic orbits with unit Poincaré determinant are excluded in dimension at least two. We also prove that if $Xh = \operatorname{div} X$, equivalently if X admits a smooth positive invariant density, then under (H) every invariant Borel probability measure is represented by a smooth density. The main dynamical consequence is that, in dimensions at most three, (H) together with this smooth-density reduction forces zero topological entropy. In particular, zero entropy follows under (H), $Xh = \operatorname{div} X$, and $\dim M \leq 3$; this includes the divergence-free case. The unconditional assertion for arbitrary divergence and arbitrary dimension is not established here, and the remaining gaps are described in the final section.

1 Introduction and relation to the original problem

The original problem asks whether hypothesis (H) alone forces the flow g_t of X to have zero topological entropy. The results below are the consequences proved by the certified arguments. They show that (H) is a strong global regularity condition, but they also identify exactly where an unconditional proof would require additional input.

Throughout, M is a compact connected smooth manifold without boundary, $n = \dim M$, X is a smooth vector field, and g_t is its flow. We fix once and for all a smooth positive density vol on M. The divergence $\operatorname{div} X$ is defined by

$$\mathcal{L}_X \mathrm{vol} = (\operatorname{div} X)\mathrm{vol}.$$

Distributions are continuous linear functionals on smooth densities. For $u \in \mathcal{D}'(M)$, the distributional Lie derivative is

$$\langle \mathcal{L}_X u, \alpha \rangle := -\langle u, \mathcal{L}_X \alpha \rangle, \qquad \alpha \in C^\infty(M, |\Lambda|).$$

A smooth function f is identified with the distribution $\alpha \mapsto \int_M f\alpha$. Under the identification of test densities with test functions by vol, one has

$$\langle \mathcal{L}_X u, \varphi \rangle = -\langle u, X\varphi + \varphi \operatorname{div} X \rangle, \qquad \varphi \in C^\infty(M),$$

where $\langle u, \varphi \rangle := \langle u, \varphi \mathrm{vol} \rangle$. For a Borel measure μ,

$$\langle \mathcal{L}_X \mu, \varphi \rangle = -\int_M (X\varphi + \varphi \operatorname{div} X)\, d\mu.$$

Hypothesis (H) is the assertion that every distributional solution of

$$\mathcal{L}_X u = f, \qquad f \in C^\infty(M),$$

is represented by a smooth function.

The main results are as follows.

- In Section 3, we prove the local flow-box normal form. Away from zeros, distributional solutions are obtained by integration along flow lines, with arbitrary transverse distributional data. Hence for $n \geq 2$ the operator is nowhere locally hypoelliptic at regular points; (H) is a global condition. For $n = 1$, the operator is elliptic away from zeros.

- In Section 4, we prove that under (H) the distributional kernel of $\mathcal{L}_X$ is exactly the finite-dimensional space of smooth first integrals, the range of $X : C^\infty(M) \to C^\infty(M)$ is closed, and smooth solvability is characterized by annihilation by the transpose kernel. The sharp converse under the canonical density convention is not asserted here; see Section 8.

- In Section 5, we prove the singular-witness exclusions. In dimension at least two, (H) excludes trace-zero equilibria and periodic orbits with unit Poincaré determinant. If X preserves a smooth positive density, then in dimension at least two there are no equilibria, no nontrivial periodic orbits, and no nonempty compact invariant null sets. In dimension one, (H) forces X to be nowhere zero, $M \cong S^1$, and the entropy is zero.

- In Section 6, we prove the integrating-factor theorem: if $Xh = \operatorname{div} X$, equivalently if there is a smooth positive invariant density, then under (H) every invariant Borel probability measure is represented by a smooth density.

- In Section 7, we prove the conditional entropy theorem: if (H) holds and every invariant Borel probability measure is a smooth density, then $h_{\mathrm{top}}(g_1) = 0$ for $\dim M \leq 3$. Combining this with the integrating-factor theorem gives zero entropy under (H), $Xh = \operatorname{div} X$, and $\dim M \leq 3$, in particular when $\operatorname{div} X = 0$.

Section 8 records the precise limitations. The full original assertion, without an integrating-factor or other smooth-density reduction and in arbitrary dimension, is not proved by the certified arguments.

2 Notation and standard facts

We use the convention established above. For a smooth function u, one has $\mathcal{L}_X u = Xu$. Indeed,

$$\langle \mathcal{L}_X u, \varphi \rangle = -\int_M u(X\varphi + \varphi \operatorname{div} X)\,\mathrm{vol} = \int_M (Xu)\varphi\,\mathrm{vol},$$

because $\int_M \mathcal{L}_X(u\varphi\mathrm{vol}) = 0$ on a compact manifold.

For a periodic orbit γ of least period $T > 0$ through p, the Poincaré determinant is

$$\det d(g_T)_p.$$

It is independent of the choice of $p \in \gamma$. We shall use Liouville's formula in the form proved in Lemma 2.1.

In this paper, the topological entropy of the flow is the topological entropy $h_{\mathrm{top}}(g_1)$ of its time-one map. This is the standard convention in smooth dynamics; proving $h_{\mathrm{top}}(g_1) = 0$ is therefore proving that the flow has zero topological entropy.

Lemma 2.1. *Let γ be a periodic orbit of least period T through p. Then*

$$\det d(g_T)_p = \exp\left(\int_0^T \operatorname{div} X(g_s p)\,ds\right).$$

Proof. Let $J_t(x)$ be the Jacobian cocycle defined by

$$(g_t)^*\mathrm{vol} = J_t\,\mathrm{vol}.$$

Differentiating,

$$\frac{d}{dt}(g_t)^*\mathrm{vol} = (g_t)^*(\mathcal{L}_X\mathrm{vol}) = (g_t)^*((\operatorname{div} X)\mathrm{vol}) = (\operatorname{div} X \circ g_t)(g_t)^*\mathrm{vol}.$$

Thus, along the orbit $g_t p$,

$$\frac{d}{dt}J_t(p) = (\operatorname{div} X)(g_t p)J_t(p), \qquad J_0(p) = 1,$$

so

$$J_T(p) = \exp\left(\int_0^T \operatorname{div} X(g_s p)\,ds\right).$$

At the period T, $g_T p = p$, and $J_T(p)$ is exactly the determinant of $d(g_T)_p$ with respect to the density vol. □

3 Flow-box normal form and local non-hypoellipticity

Proposition 3.1. *Let $p \in M$ with $X(p) \neq 0$. There are local coordinates $(t, y) \in I \times D$ near p, with $I \subset \mathbb{R}$ an interval and $D \subset \mathbb{R}^{n-1}$ a ball (a point if $n = 1$), such that $X = \partial_t$. For $f \in C^\infty(I \times D)$ and $u \in \mathcal{D}'(I \times D)$,*

$$\mathcal{L}_X u = f$$

if and only if

$$u(t,y) = \int_0^t f(s,y)\,ds + c(y)$$

for a unique $c \in \mathcal{D}'(D)$ *(for* $n = 1$*,* c *is a constant). Consequently:*

1. $u \in C^\infty$ *if and only if* $c \in C^\infty$*.*
2. *If* $n \geq 2$*, then* $\mathcal{L}_X$ *is not hypoelliptic at any regular point: the distribution* $u_0(t,y) = \delta_0(y)$ *satisfies* $\mathcal{L}_X u_0 = 0$ *but is not smooth.*
3. *If* $n = 1$*, then at every point with* $X \neq 0$ *the operator is elliptic, hence hypoelliptic.*

Thus for $n \geq 2$*, hypothesis (H) is not a local condition at regular points; it is a global recurrence/gluing condition. For* $n = 1$*, the content of (H) is concentrated at the zeros of* X*.*

Proof. By the flow-box theorem (straightening theorem for vector fields; see Lee [1, Chapter 9]), there are coordinates near p in which $X = \partial_t$. Using the coordinate density, for which $\mathcal{L}_{\partial_t}$ annihilates the density, the distributional Lie derivative is simply ∂_t.

First solve the homogeneous equation $\partial_t u = 0$. Choose $\rho \in C_c^\infty(I)$ with

$$\int_I \rho(t)\,dt = 1.$$

Define $c \in \mathcal{D}'(D)$ by

$$\langle c, \eta\rangle := \langle u, \rho \otimes \eta\rangle, \qquad \eta \in C_c^\infty(D).$$

Let $\varphi \in C_c^\infty(I \times D)$, and put

$$\psi(y) := \int_I \varphi(s,y)\,ds.$$

Define

$$\Phi(t,y) := \int_{-\infty}^t \big(\varphi(s,y) - \rho(s)\psi(y)\big)\,ds.$$

Because

$$\int_I \big(\varphi(s,y) - \rho(s)\psi(y)\big)\,ds = \psi(y) - \psi(y) = 0,$$

the function Φ has compact support in $I \times D$, and

$$\varphi = \partial_t \Phi + \rho \otimes \psi.$$

Since $\partial_t u = 0$,

$$\langle u, \varphi\rangle = \langle u, \partial_t \Phi\rangle + \langle u, \rho \otimes \psi\rangle = \langle c, \psi\rangle.$$

Thus $u = c(y)$. Uniqueness follows by testing with $\rho \otimes \eta$.

For the inhomogeneous equation, define

$$F(t,y) := \int_0^t f(s,y)\,ds.$$

Then $F \in C^\infty$ and $\partial_t F = f$. Hence $\partial_t u = f$ if and only if $\partial_t(u - F) = 0$, so

$$u = F + c(y)$$

with unique $c \in \mathcal{D}'(D)$. Since F is smooth, u is smooth if and only if c is smooth.

If $n \geq 2$, choose coordinates so that $0 \in D$. Define

$$\langle u_0, \varphi \rangle := \int_I \varphi(t, 0)\, dt.$$

Then $u_0 = \delta_0(y)$, and

$$\langle \partial_t u_0, \varphi \rangle = -\int_I \partial_t \varphi(t, 0)\, dt = 0$$

because φ has compact support in I. Thus $\mathcal{L}_X u_0 = 0 \in C^\infty$, but u_0 is singular along $\{y = 0\}$. Therefore $\mathcal{L}_X$ is not hypoelliptic at p. Since p was an arbitrary regular point, it is nowhere locally hypoelliptic on $M \setminus \{X = 0\}$ when $n \geq 2$.

If $n = 1$, locally $X = a(x)\partial_x$ with $a \neq 0$. If $\mathcal{L}_X u = f \in C^\infty$, then

$$a\, \partial_x u = f,$$

so $\partial_x u = f/a \in C^\infty$. A distribution with smooth derivative is locally a smooth function plus a constant, hence u is smooth. Thus $\mathcal{L}_X$ is elliptic, hence hypoelliptic, at regular points in dimension one. □

4 Functional-analytic structure of hypothesis (H)

We say that X is C^∞-stable if the range

$$X(C^\infty(M)) \subset C^\infty(M)$$

is closed in the Fréchet topology of $C^\infty(M)$.

Theorem 4.1. *Assume (H). Then:*

1. $\ker(\mathcal{L}_X : \mathcal{D}'(M) \to \mathcal{D}'(M)) = \{u \in C^\infty(M) : Xu = 0\}$, *and this space is finite-dimensional.*

2. $X : C^\infty(M) \to C^\infty(M)$ *has closed range; equivalently, X is C^∞-stable.*

3. *For $f \in C^\infty(M)$, the equation $Xh = f$ has a smooth solution $h \in C^\infty(M)$ if and only if*

$$\ell(f) = 0$$

for every ℓ in the kernel of the transpose of $X : C^\infty(M) \to C^\infty(M)$ on the continuous dual of $C^\infty(M)$. Under the identification of distributions with linear functionals on smooth functions via vol*, this says that f must be annihilated by every distribution T satisfying*

$$\langle T, (X\varphi)\mathrm{vol} \rangle = 0 \qquad \text{for all } \varphi \in C^\infty(M).$$

Proof. Write $S = \mathcal{L}_X$. Fix a Riemannian metric and let $H^s(M)$ denote the usual Sobolev spaces of distributions, defined using the fixed density vol. We use the Rellich–Kondrachov compactness theorem on compact manifolds: if $s > t$, then the inclusion $H^s(M) \hookrightarrow H^t(M)$ is compact; see Evans [2, Theorem 5.7.1]. The distributional operator S is first order, hence maps H^{-m} continuously into H^{-m-1}.

For each integer $m \geq 0$, define

$$F_m := \{u \in H^{-m}(M) : Su \in C^\infty(M)\},$$

with the Fréchet topology given by the seminorms

$$p_{m,k}(u) := \|u\|_{H^{-m}} + \|Su\|_{C^k}, \qquad k = 0, 1, 2, \ldots.$$

If u_j is Cauchy in F_m, then $u_j \to u$ in H^{-m} and $Su_j \to g$ in C^∞. Since $S : H^{-m} \to H^{-m-1}$ is continuous, $Su_j \to Su$ in H^{-m-1}, hence $Su = g \in C^\infty$. Thus $u \in F_m$, and F_m is a Fréchet space.

By (H), as sets $F_m = C^\infty(M)$. The identity

$$\iota_m : F_m \to C^\infty(M)$$

is therefore bijective. Its graph is closed: if $u_j \to u$ in F_m and $u_j \to v$ in C^∞, then both limits agree in H^{-m}, so $u = v$. By the closed graph theorem for Fréchet spaces, ι_m is continuous. In particular, for the continuous seminorm $\|\cdot\|_{C^0}$ on C^∞, there exist constants $C_m > 0$ and $k_m \geq 0$ such that

$$\|u\|_{C^0} \leq C_m\big(\|Su\|_{C^{k_m}} + \|u\|_{H^{-m}}\big), \qquad u \in C^\infty(M).$$

We shall use the cases $m = 1$ and $m = 2$:

$$\|u\|_{C^0} \leq C_1\big(\|Su\|_{C^{k_1}} + \|u\|_{H^{-1}}\big), \tag{1}$$

$$\|u\|_{C^0} \leq C_2\big(\|Su\|_{C^{k_2}} + \|u\|_{H^{-2}}\big). \tag{2}$$

Let

$$K := \ker(S : \mathcal{D}'(M) \to \mathcal{D}'(M)).$$

By (H), every element of K is smooth; since smooth solutions of $Su = 0$ satisfy $Xu = 0$, we have

$$K = \{u \in C^\infty(M) : Xu = 0\}.$$

We prove K is finite-dimensional. Let

$$B := \{u \in K : \|u\|_{C^0} \leq 1\}.$$

Since $C^0 \hookrightarrow H^{-1}$ continuously, B is bounded in H^{-1}. By Rellich compactness, B is pre-compact in H^{-2}. Applying (2) to differences of elements of B, and using $Su = 0$ on K, gives

$$\|u - v\|_{C^0} \leq C_2\|u - v\|_{H^{-2}}, \qquad u, v \in K.$$

Thus B is precompact in C^0. A normed vector space whose unit ball is precompact is finite-dimensional, so $\dim K < \infty$.

We next prove closedness of $X(C^\infty)$. Let $f_j = Xh_j \to f$ in C^∞, with $h_j \in C^\infty$. Since K is finite-dimensional, choose a closed complement W of K in C^∞. More explicitly, choose a basis $q_1, \dots, q_r$ of K and smooth functions $r_1, \dots, r_r$ with

$$\int_M q_i r_j \,\mathrm{vol} = \delta_{ij},$$

and set

$$W := \left\{ h \in C^\infty(M) : \int h r_i \,\mathrm{vol} = 0,\ i = 1, \dots, r \right\}.$$

The functionals $h \mapsto \int h r_i \,\mathrm{vol}$ are continuous in the C^0-norm, so W is closed in C^0. Replacing each h_j by

$$h_j - \sum_i \left(\int h_j r_i \,\mathrm{vol} \right) q_i,$$

we may assume $h_j \in W$.

We first show that (h_j) is bounded in C^0. If not, after passing to a subsequence,

$$a_j := \|h_j\|_{C^0} \to \infty.$$

Set $v_j := h_j / a_j$. Then $\|v_j\|_{C^0} = 1$, $v_j \in W$, and

$$Sv_j = f_j / a_j \to 0$$

in C^∞. The sequence (v_j) is bounded in C^0, hence bounded in L^2, hence bounded in H^0. By Rellich compactness $H^0 \hookrightarrow H^{-1}$, a subsequence is Cauchy in H^{-1}, hence Cauchy in H^{-2}. Applying (2) to differences $v_j - v_\ell$, we obtain

$$\|v_j - v_\ell\|_{C^0} \le C_2 \big(\|S(v_j - v_\ell)\|_{C^{k_2}} + \|v_j - v_\ell\|_{H^{-2}} \big) \to 0.$$

Thus (v_j) is Cauchy in C^0. Let $v_j \to v$ in C^0. Since $Sv_j \to 0$ in distributions and $v_j \to v$ in distributions, continuity of S on distributions gives $Sv = 0$. By (H), $v \in C^\infty$, hence $v \in K$. But W is closed under C^0-limits, so $v \in W \cap K = \{0\}$, contradicting $\|v\|_{C^0} = 1$. Therefore (h_j) is bounded in C^0.

Being bounded in C^0, (h_j) is bounded in L^2. By Rellich compactness $H^0 \hookrightarrow H^{-1}$, a subsequence converges in H^{-1}, hence is Cauchy in H^{-1}. Since (f_j) is Cauchy in C^{k_1}, estimate (1) applied to differences $h_j - h_\ell$ gives

$$\|h_j - h_\ell\|_{C^0} \le C_1 \big(\|f_j - f_\ell\|_{C^{k_1}} + \|h_j - h_\ell\|_{H^{-1}} \big) \to 0.$$

Thus (h_j) is Cauchy in C^0, converging to some $h \in C^0$. Then $Sh = f$ in distributions. Since f is smooth, (H) gives $h \in C^\infty$. Thus $f = Xh \in X(C^\infty)$, proving that the range is closed.

For the solvability criterion, since $X : C^\infty \to C^\infty$ is continuous with closed range, the Hahn–Banach closed-range theorem for Fréchet spaces gives

$$X(C^\infty) = \{ f \in C^\infty(M) : \ell(f) = 0 \text{ for every } \ell \in \ker X^t \},$$

where X^t is the transpose on the continuous dual of $C^\infty(M)$; see Trèves [3, Theorem 37.2]. Under the density identification, ℓ is represented by a distribution T, and $\ell \in \ker X^t$ means precisely

$$\langle T, (X\varphi)\mathrm{vol}\rangle = 0 \qquad \forall \varphi \in C^\infty(M).$$

This proves the criterion. □

5 Zeros, periodic orbits, and invariant smooth densities

Theorem 5.1. *Assume (H), with the canonical density convention. Let $n = \dim M$.*

1. *If $n \geq 2$, then X has no periodic orbit γ of least period $T > 0$ with*

$$\det d(g_T)_p = 1$$

for one (hence every) $p \in \gamma$.

2. *If $n \geq 2$, then X has no zero p with*

$$\operatorname{tr} DX(p) = 0.$$

3. *If, in addition, X preserves a smooth positive density ν, i.e.*

$$\mathcal{L}_X \nu = 0,$$

and $n \geq 2$, then X has no zeros, no nontrivial periodic orbits, and every nonempty compact g_t-invariant set has positive ν-measure.

4. *If $n = 1$, then (H) implies that X is nowhere zero. Hence $M \cong S^1$, every orbit is the whole circle, every nowhere-zero smooth vector field on S^1 satisfies (H), and*

$$h_{\mathrm{top}}(g_1) = 0.$$

Proof. We prove the four items in order.

(1) Periodic orbit with unit Poincaré determinant. Let $\gamma = \{g_s p : 0 \leq s < T\}$ be a periodic orbit of least period T, with $n \geq 2$. Fix the background density vol, and write

$$D(s) := \int_0^s \operatorname{div} X(g_r p)\, dr.$$

By Liouville's formula, Lemma 2.1,

$$\det d(g_T)_p = e^{D(T)}.$$

Assume $\det d(g_T)_p = 1$, so $D(T) = 0$. Then

$$\widehat{h}(s) := e^{D(s)}$$

is a well-defined smooth positive function on $\gamma \cong \mathbb{R}/T\mathbb{Z}$, because

$$\widehat{h}(s+T) = e^{D(s)+D(T)} = \widehat{h}(s).$$

Since γ is a closed embedded submanifold, $\log \widehat{h}$ extends to a smooth function on M; exponentiating gives a smooth function $c > 0$ on all of M such that $c(g_s p) = \widehat{h}(s)$.

Let μ_γ be the normalized time-average on γ:

$$\mu_\gamma := \frac{1}{T} \int_0^T \delta_{g_s p}\, ds.$$

Define a distribution T_γ by

$$\langle T_\gamma, \varphi \mathrm{vol} \rangle := \int_M \varphi\, c\, d\mu_\gamma = \frac{1}{T} \int_0^T \varphi(g_s p) c(g_s p)\, ds.$$

For $\varphi \in C^\infty(M)$,

$$\begin{aligned} \langle \mathcal{L}_X T_\gamma, \varphi \mathrm{vol} \rangle &= -\langle T_\gamma, \mathcal{L}_X(\varphi \mathrm{vol}) \rangle \\ &= -\frac{1}{T} \int_0^T \big(X\varphi(g_s p) c(g_s p) + \varphi(g_s p) c(g_s p) \operatorname{div} X(g_s p) \big)\, ds. \end{aligned}$$

Along γ,

$$\frac{d}{ds} c(g_s p) = c(g_s p) \operatorname{div} X(g_s p),$$

so the integrand is

$$\frac{d}{ds} \big(\varphi(g_s p) c(g_s p) \big).$$

Its integral over one period is zero. Hence

$$\mathcal{L}_X T_\gamma = 0.$$

By (H), T_γ is represented by a smooth function q:

$$T_\gamma = q\, \mathrm{vol}.$$

But T_γ is supported on γ. Since $n \geq 2$, the embedded circle γ has vol-measure zero and empty interior. Therefore $q = 0$ on the dense open set $M \setminus \gamma$, hence $q \equiv 0$. On the other hand, because $c > 0$,

$$\langle T_\gamma, c^{-1} \mathrm{vol} \rangle = \int_M c^{-1} c\, d\mu_\gamma = 1,$$

a contradiction. Thus no such periodic orbit exists.

(2) Trace-zero zero. Let $p \in M$ with $X(p) = 0$. Define the Dirac distribution δ_p by

$$\langle \delta_p, \varphi \mathrm{vol} \rangle := \varphi(p).$$

Then

$$\langle \mathcal{L}_X \delta_p, \varphi \mathrm{vol} \rangle = -(X\varphi)(p) - \varphi(p) \operatorname{div} X(p).$$

Since $X(p) = 0$, $(X\varphi)(p) = 0$. At a zero, $\operatorname{div} X(p) = \operatorname{tr} DX(p)$. Hence

$$\langle \mathcal{L}_X \delta_p, \varphi \mathrm{vol} \rangle = -\varphi(p) \operatorname{tr} DX(p).$$

If $\operatorname{tr} DX(p) = 0$, then $\mathcal{L}_X \delta_p = 0$. By (H), $\delta_p = q\,\mathrm{vol}$ for some smooth q. Since $\{p\}$ has empty interior and vol-measure zero, $q \equiv 0$, but $\langle \delta_p, \mathrm{vol} \rangle = 1$, contradiction. Thus no such zero exists when $n \geq 2$.

(3) Existence of an invariant smooth positive density. Assume $\mathcal{L}_X \nu = 0$, with ν smooth and positive. Let μ be a finite g_t-invariant Borel measure. For $\varphi \in C^\infty(M)$, invariance gives

$$\int_M \varphi \circ g_t \, d\mu = \int_M \varphi \, d\mu.$$

Differentiating at $t = 0$ yields

$$\int_M X\varphi \, d\mu = 0.$$

Using $\mathcal{L}_X \nu = 0$,

$$\langle \mathcal{L}_X \mu, \varphi\nu \rangle = -\langle \mu, \mathcal{L}_X(\varphi\nu) \rangle = -\langle \mu, (X\varphi)\nu \rangle = -\int_M X\varphi \, d\mu = 0.$$

Thus $\mathcal{L}_X \mu = 0$. By (H), $\mu = q\nu$ for some $q \in C^\infty(M)$. If $\operatorname{supp} \mu$ is ν-null, then $q = 0$ on the dense open set $M \setminus \operatorname{supp} \mu$, hence $q \equiv 0$, so $\mu = 0$.

If p is a zero of X, then δ_p is invariant, so $\delta_p = q\nu$. Since $n \geq 2$, $\nu(\{p\}) = 0$, contradiction. If γ is a nontrivial periodic orbit, its time-average μ_γ is invariant and supported on an embedded circle, which is ν-null for $n \geq 2$, again a contradiction.

Finally, let $K \neq \varnothing$ be compact and g_t-invariant. Choose $x \in K$, and define Krylov–Bogolyubov averages

$$\mu_m := \frac{1}{m} \int_0^m \delta_{g_t x} \, dt.$$

Since K is compact and invariant, every weak-* limit μ of a subsequence is a g_t-invariant Borel probability supported on K; see Walters [4, Chapter 6]. For completeness, we recall the invariance check. For $\varphi \in C(M)$ and $s \geq 0$,

$$\left| \int \varphi \circ g_s \, d\mu_m - \int \varphi \, d\mu_m \right| = \frac{1}{m} \left| \int_m^{m+s} \varphi(g_t x) \, dt - \int_0^s \varphi(g_t x) \, dt \right| \leq \frac{2s\|\varphi\|_\infty}{m} \to 0.$$

The case $s < 0$ is analogous. Differentiating at $s = 0$ for smooth φ gives $\int X\varphi \, d\mu = 0$. Hence, by the preceding paragraph, $\mu = q\nu$. If $\nu(K) = 0$, then μ is supported on a ν-null set, so $\mu = 0$, contradicting $\mu(M) = 1$. Therefore $\nu(K) > 0$.

(4) Dimension one. If $n = 1$, then $M \cong S^1$. Choose a coordinate θ and write

$$X = c(\theta)\partial_\theta.$$

We first show that (H) forces c to be nowhere zero. Suppose c has a zero. If there is a zero p with $c(p) = c'(p) = 0$, then $\operatorname{tr} DX(p) = c'(p) = 0$, and the Dirac witness above contradicts

(H). If every zero is simple, then the zeros of c on S^1 are isolated and occur in pairs, with sign changes. Hence $S^1 \setminus Z(c)$ has a nonempty proper open component I whose boundary consists of zeros of c. Let χ_I be the characteristic function of I, regarded as a distribution. With respect to the density $d\theta$,

$$\langle \mathcal{L}_X \chi_I, \varphi d\theta \rangle = -\int_I (c\varphi)'\, d\theta = -[c\varphi]_{\partial I} = 0.$$

But χ_I is not represented by a smooth function, contradicting (H). Therefore c has no zeros.

Assume after reversing orientation that $c > 0$. We show that (H) holds for every such vector field. Let $u \in \mathcal{D}'(S^1)$ and suppose $\mathcal{L}_X u = f \in C^\infty$. With the density $d\theta$,

$$\langle \mathcal{L}_X u, \varphi d\theta \rangle = -\langle u, (c\varphi)' d\theta \rangle = \int f\varphi\, d\theta.$$

Since multiplication by c is an automorphism of $C^\infty(S^1)$, writing $\psi = c\varphi$ gives

$$-\langle u, \psi' \rangle = \int (f/c)\psi\, d\theta.$$

Thus $u' = f/c$ in distributions. Since f/c is smooth, u is a constant plus a smooth primitive; hence u is smooth.

Since $c > 0$, every orbit is the whole circle. The period is

$$T = \int_{S^1} \frac{d\theta}{c(\theta)} < \infty.$$

Define

$$\Phi(\theta) := \frac{1}{T} \int_0^\theta \frac{ds}{c(s)} \pmod 1.$$

Then $\Phi : S^1 \to S^1$ is a smooth diffeomorphism and

$$\Phi_*(X) = \frac{1}{T}\, \partial_\tau.$$

Thus g_1 is smoothly conjugate to the rotation $\tau \mapsto \tau + 1/T$ on S^1. A rotation is an isometry for the standard metric, so its topological entropy is zero; topological entropy is invariant under conjugacy. Hence

$$h_{\mathrm{top}}(g_1) = 0.$$

□

6 Integrating factors and smoothness of invariant measures

Theorem 6.1. *Assume (H). Fix the smooth positive density* vol*, and suppose there exists* $h \in C^\infty(M)$ *such that*

$$Xh = \operatorname{div} X.$$

Equivalently, $e^{-h}\mathrm{vol}$ is a smooth positive X-invariant density. Then every g_t-invariant Borel probability measure μ is represented by a smooth density:

$$\mu = \rho\,\mathrm{vol}, \qquad \rho \in C^\infty(M), \quad \rho \geq 0.$$

In particular, the conclusion holds unconditionally when $\operatorname{div} X \equiv 0$ *(take $h \equiv 0$).*

Proof. Let $w := e^h > 0$. Then

$$Xw = w\,\operatorname{div} X.$$

Also,

$$\mathcal{L}_X(e^{-h}\mathrm{vol}) = \big(-e^{-h}Xh + e^{-h}\operatorname{div} X\big)\mathrm{vol} = 0,$$

so $e^{-h}\mathrm{vol}$ is invariant.

Let μ be a g_t-invariant Borel probability measure. For every smooth function ψ,

$$\int_M X\psi\,d\mu = 0,$$

by differentiating $\int \psi \circ g_t\,d\mu = \int \psi\,d\mu$ at $t = 0$.

Consider the measure $w\mu$. For $\varphi \in C^\infty(M)$,

$$\begin{aligned}
\langle \mathcal{L}_X(w\mu), \varphi\mathrm{vol}\rangle &= -\int_M (X\varphi + \varphi \operatorname{div} X)\, w\,d\mu \\
&= -\int_M \big(wX\varphi + \varphi w \operatorname{div} X\big)\,d\mu \\
&= -\int_M \big(wX\varphi + \varphi Xw\big)\,d\mu \\
&= -\int_M X(w\varphi)\,d\mu \\
&= 0.
\end{aligned}$$

Thus $\mathcal{L}_X(w\mu) = 0$. By (H), there exists $q \in C^\infty(M)$ such that

$$w\mu = q\,\mathrm{vol}.$$

Since $w > 0$,

$$\mu = \frac{q}{w}\,\mathrm{vol}.$$

Set $\rho := q/w$. Because μ is a positive measure and $w > 0$, the smooth function q is nonnegative vol-almost everywhere, hence everywhere by continuity. Therefore $\rho \geq 0$.

If $\operatorname{div} X \equiv 0$, take $h \equiv 0$, so $w = 1$, and the same argument applies. □

7 Zero entropy in dimensions at most three under the smooth-density reduction

Theorem 7.1. *Assume (H), and assume further that every g_t-invariant Borel probability measure on M is represented by a smooth density with respect to* vol. *If*

$$\dim M \leq 3,$$

then

$$h_{\mathrm{top}}(g_1) = 0.$$

Thus the flow has zero topological entropy.

Proof. If $\dim M = 0$, then M is a point and the conclusion is trivial. Assume $n = \dim M \geq 1$.

First observe that the smooth-density hypothesis implies that X has no zeros. Indeed, if $X(p) = 0$, then δ_p is a g_t-invariant Borel probability measure, but a smooth density cannot be supported at a single point.

Suppose, for contradiction, that

$$h_{\mathrm{top}}(g_1) > 0.$$

By the variational principle for the continuous map g_1 on the compact space M, there exists a g_1-invariant Borel probability measure ν with

$$h_\nu(g_1) > 0.$$

Average ν over one unit of flow time:

$$\bar{\nu} := \int_0^1 (g_s)_*\nu \, ds.$$

Then $\bar{\nu}$ is g_t-invariant. Since each g_s commutes with g_1, the systems (M, ν, g_1) and $(M, (g_s)_*\nu, g_1)$ are measurably conjugate, so

$$h_{(g_s)_*\nu}(g_1) = h_\nu(g_1).$$

By affinity of metric entropy for invariant measures, $h_{\bar{\nu}}(g_1) = h_\nu(g_1) > 0$. Using the ergodic decomposition for the flow and affinity of entropy, we obtain a g_t-invariant ergodic Borel probability measure μ with

$$h_\mu(g_1) > 0.$$

These standard facts are in Walters [4, Chapters 6 and 7].

Apply Oseledets' multiplicative ergodic theorem [5] to the derivative cocycle dg_1. Let the Lyapunov exponents of μ be

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n.$$

Because X is nowhere zero, the line spanned by X is a smooth invariant line. For $m \geq 1$,

$$d(g_1)^m X(x) = X(g_m x),$$

and since $\|X\|$ is bounded above and below away from zero, the Lyapunov exponent of this line is zero. Thus one of the λ_i is 0.

Ruelle's inequality [6] applied to the C^∞ diffeomorphism g_1 gives

$$h_\mu(g_1) \leq \sum_{\lambda_i > 0} \lambda_i.$$

Since $h_\mu(g_1) > 0$, there is at least one strictly positive Lyapunov exponent. Metric entropy is invariant under taking inverses for invertible transformations, so $h_\mu(g_{-1}) = h_\mu(g_1) > 0$.

Applying Ruelle's inequality to g_{-1}, whose Lyapunov exponents are $-\lambda_i$, shows that there is at least one strictly negative Lyapunov exponent for g_1.

If $n = 1$, the only exponent is the flow direction exponent 0, contradicting the existence of a positive exponent. If $n = 2$, the two exponents would have to include 0, a positive exponent, and a negative exponent, which is impossible. Hence for $n \leq 2$ we obtain a contradiction.

Now assume $n = 3$. The Oseledets spectrum of μ must be exactly

$$\chi^- < 0, \qquad 0, \qquad \chi^+ > 0,$$

where the zero exponent is the flow direction. Thus μ is a hyperbolic invariant measure for the flow, with no transverse zero exponents.

Since g_t is C^∞, in particular $C^{1+\alpha}$, Katok's approximation theorem for flows applies. In the form needed here [7, Section 4]: if a $C^{1+\alpha}$ flow has an ergodic hyperbolic invariant probability measure with positive metric entropy, then for every $\varepsilon > 0$ there exists a compact invariant locally maximal uniformly hyperbolic set Λ such that

$$h_{\mathrm{top}}(g_1|_\Lambda) > h_\mu(g_1) - \varepsilon.$$

Choosing $0 < \varepsilon < h_\mu(g_1)$, we obtain a flow horseshoe with positive topological entropy. By its construction as a suspension over a topological Markov section, Λ contains a nontrivial periodic flow orbit γ.

Let ν_γ be the normalized time-average measure on γ. It is a g_t-invariant Borel probability measure. Since $n = 3$, the orbit γ is an embedded circle and has vol-measure zero. Hence ν_γ is singular with respect to vol. This contradicts the hypothesis that every invariant Borel probability measure is represented by a smooth density. Therefore $h_{\mathrm{top}}(g_1) = 0$. □

Combining Theorem 7.1 with Theorem 6.1 gives the main low-dimensional entropy corollary.

Corollary 7.2. *Assume (H). If there exists $h \in C^\infty(M)$ such that*

$$Xh = \operatorname{div} X,$$

and if

$$\dim M \leq 3,$$

then

$$h_{\mathrm{top}}(g_1) = 0.$$

Thus the flow has zero topological entropy. In particular, this holds when $\operatorname{div} X \equiv 0$.

Proof. By Theorem 6.1, every invariant Borel probability measure is represented by a smooth density. The conclusion then follows from Theorem 7.1. □

8 Open points and limitations

The certified results above do not prove the original assertion in full generality. The remaining gaps are as follows.

1. **Invariant measures and the divergence term.** For a general invariant Borel probability measure μ,
$$\mathcal{L}_X \mu = -(\operatorname{div} X)\mu$$
as distributions. Thus applying (H) directly to μ requires either $\operatorname{div} X = 0$ on the support or an integrating factor $Xh = \operatorname{div} X$. The smooth-density reduction is proved here only under that integrating-factor hypothesis.

2. **The sharp functional-analytic converse.** The forward implications of Theorem 4.1 are proved. The converse statement that C^∞-stability together with smoothness of $\ker(\mathcal{L}_X : \mathcal{D}' \to \mathcal{D}')$ implies (H) is not asserted under the canonical density convention. The annihilator relevant to the closed range of $X : C^\infty \to C^\infty$ is the kernel of the transpose of X on functionals on functions, whereas the canonical kernel $\ker \mathcal{L}_X$ is a different object unless additional hypotheses are imposed.

3. **Residual cohomological obstructions.** The elementary Dirac and weighted orbit-current witnesses exclude trace-zero equilibria and periodic orbits with unit Poincaré determinant. They do not exclude zeros with nonzero trace or periodic orbits with non-unit Poincaré determinant. The remaining obstruction is cohomological in nature and is measured by the Livšic/de Rham class of $\operatorname{div} X$, equivalently by period integrals of $\operatorname{div} X$ over closed orbits.

4. **Dimension four and higher.** In dimensions at least four, an ergodic positive-entropy measure may have additional zero Lyapunov exponents transverse to the flow. Katok's horseshoe theorem applies to hyperbolic measures; when transverse zero exponents are present, the certified arguments do not supply a replacement.

5. **Absence of zeros and periodic orbits is insufficient.** The absence of zeros and periodic orbits alone does not force zero entropy. Positive-entropy smooth flows without periodic orbits exist, for example constant-roof suspensions of products of a hyperbolic toral automorphism with an irrational rotation. Thus genuine rigidity or production of singular invariant distributions is needed, not merely orbit exclusion.

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